THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VIII — MATHEMATICS · MID-TERM PRACTICE PAPER

Session 2026-27 | CBSE / RBSE Pattern

Class: 8	Subject: Mathematics	Set 01	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. This question paper contains 16 questions. All questions are compulsory.
2. Question paper is divided into FIVE sections – Section 'A', 'B', 'C', 'D' and 'E'.
3. In Section 'A' question number 1 has 12 multiple choice sub-parts carrying 1 mark each.
4. In Section 'B' question numbers 2 to 7 are very short answer type questions carrying 2 marks each.
5. In Section 'C' question numbers 8 to 10 are short answer (SA) type questions carrying 3 marks each.
6. In Section 'D' question numbers 11 to 13 are long answer (LA) type questions carrying 5 marks each.
7. In Section 'E' question numbers 14 to 16 are case-based questions whose sub-parts are of 1, 1 and 2 marks respectively.
8. There is no overall choice. However, internal choice has been provided in 1 question of Section B, 1 question of Section C, 2 questions of Section D, and in the 2-mark sub-part of every case-based question in Section E. Attempt only one of the alternatives in all such questions.
9. Draw neat figures wherever required. Use of calculators is NOT allowed. Write the serial number of the question before attempting it.

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) The smallest 4-digit perfect square number is: **1**

(a) 1000 (b) 1089 (c) 1024 (d) 1156

(ii) The cube root of 17576 is: **1**

(a) 24 (b) 28 (c) 26 (d) 36

(iii) The standard form of 0.00045 is: **1**

(a) 4.5×10^{-3} (b) 45×10^{-5} (c) 4.5×10^{4} (d) 4.5×10^{-4}

(iv) The Roman numeral XCIV represents: **1**

(a) 86 (b) 94 (c) 106 (d) 114

(v) The binary number 1011 is equal to which decimal number? **1**

(a) 11 (b) 9 (c) 10 (d) 13

(vi) Which quadrilateral always has equal diagonals that bisect each other, but the diagonals need not be perpendicular? **1**

(a) Rhombus (b) Kite (c) Rectangle (d) Parallelogram

(vii) A number is divisible by 18 if it is divisible by: **1**

(a) 2 and 9 (b) 3 and 6 (c) 4 and 9 (d) 2 and 6

(viii) The value of $104^2 - 96^2$ is: **1**

(a) 800 (b) 1600 (c) 1200 (d) 16

(ix) In the expansion of $(7x - 3)^2$, the middle term is: **1**

(a) $-42x$ (b) $-21x$ (c) $+42x$ (d) $-9x$

(x) Which of the following pairs of ratios forms a proportion? **1**

(a) 15 : 25 and 9 : 15 (b) 4 : 6 and 9 : 12

(c) 3 : 5 and 5 : 3 (d) 2 : 7 and 7 : 2

(xi) The ratio of 45 minutes to 1½ hours in simplest form is: **1**

(a) 3 : 2 (b) 1 : 2 (c) 1 : 3 (d) 2 : 3

(xii) Sum of any two consecutive odd numbers is always divisible by: **1**

(a) 3 (b) 4 (c) 5 (d) 6

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the square root of 2916 by prime factorisation. **2**

3. Write 0.000000837 in standard form. **2**

4. Write 2024 in Roman numerals. **2**

5. Convert the decimal number 45 into base 2. **2**

OR

Find the decimal value of (2314) in base 5.

6. Two adjacent angles of a parallelogram are in the ratio 2 : 3. Find all the angles of the parallelogram. **2**

7. Find x if x : 12 :: 15 : 36. **2**

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Find the sum without converting to Hindu-Arabic numerals: **3**
(i) XLVII + LXIII (ii) CCXLIII + DCLVII

9. The diagonals of a rhombus are 16 cm and 30 cm. Find the length of its side and its perimeter. **3**

OR

In parallelogram PQRS, $\angle P = (3x + 10)°$ and $\angle Q = (2x - 10)°$. Find x and all the angles.

10. Expand and simplify: **3**
(i) $(x + 4)(x^2 - 2x + 3)$ (ii) $(2m - 1)(m + 3)$

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 5292 must be multiplied so that the product is a perfect square. Also find the square root of the product. **5**
(ii) Find the side of a square whose area is 1296 m².

OR

(i) Find the smallest number by which 3087 must be divided so that the quotient is a perfect cube. Also find the cube root of the quotient.
(ii) The volume of a cube is 5832 cm³. Find the length of its edge.

12. If $13^2 = 169$, find the values of: (i) 1.3^2 (ii) 0.13^2 (iii) 130^2 (iv) 1300^2 (v) 0.013^2 **5**

OR

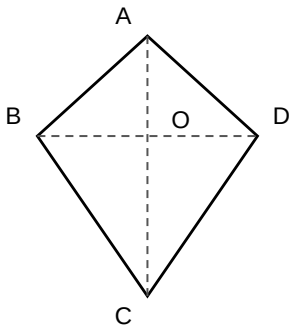
Simplify and write in exponential form: (i) $7^{-3} \times 7^{-2}$ (ii) $3^9 \div 3^5$ (iii) $(5^{-2})^{-3}$ (iv) $2^7 \div 2^{-3}$ (v) $a^4b^6 \times (ab)^3$

13. (i) Find the digit(s) that can replace * in 52*8 so that the number is divisible by 3. **5**
(ii) Is 3141592 divisible by 11? Show your work.
(iii) Find the smallest 3-digit number divisible by 4 and 9 both.
(iv) Replace the blank to make 7_6 divisible by 4 (smallest digit).
(v) A number is divisible by 6. Name its two divisibility conditions.

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. Riya and Aarav are making a kite for the school exhibition. The kite ABCD has two pairs of adjacent equal sides: AB = AD = 6 cm and CB = CD = 9 cm. Its diagonals AC and BD meet at O (see figure).



(i) What is the length of the side CD? 1

(ii) Name the pair of adjacent angles of the kite that are equal, and state the reason. 1

(iii) Show that the diagonal AC bisects ∠A and ∠C. 2

OR

(iii) If ∠AOB = x°, find x and give a reason.

15. Aditi and Rahul prepare for a Mental Maths Quiz. Their teacher shows the identities: (A) $(a + b)^2 = a^2 + 2ab + b^2$; (B) $(a - b)^2 = a^2 - 2ab + b^2$; (C) $a^2 - b^2 = (a + b)(a - b)$.

(i) Find 53^2 using identity (A). 1

(ii) Find 1.02×0.98 using identity (C). 1

(iii) Prove that $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$. 2

OR

(iii) Find $1003^2 - 997^2$ using a suitable identity.

16. The school garden club prepares a fertiliser mixture of urea, potash and compost in the ratio 2 : 3 : 5. The total weight of the mixture is 40 kg.

(i) How many kilograms of compost are in the mixture? 1

(ii) Write the ratio of urea to compost in simplest form. 1

(iii) The club uses 7.5 kg of potash in another batch of the same mixture. What is the total weight of that batch? 2

OR

(iii) Find the missing numbers so that these ratios are proportional to 2 : 5: 14 : __ and __ : 45